11.3 The Dot Product

Given two vectors $\mathbf{u} =< u_1, u_2, u_3 >$ and $\mathbf{v} =< v_1, v_2, v_3 >$ in $\mathbb{R}^3$, the **dot product** of the vectors $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u} \cdot \mathbf{v}$, is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

In the two dimensional space, if $\mathbf{u} =< u_1, u_2 >$ and $\mathbf{v} =< v_1, v_2 >$, then

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

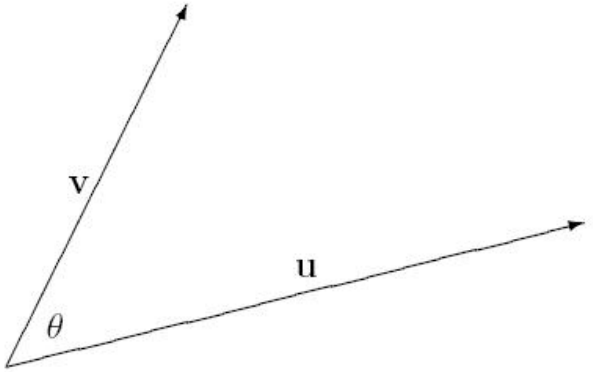
Example:

1. $\mathbf{u} =< 1, 2, 3 >, \mathbf{v} =< 3, 2, -1 >$

2. $\mathbf{u} = \frac{1}{2}\mathbf{i} + 3\mathbf{j} + \mathbf{k}, \mathbf{v} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

3. $\mathbf{u} =< \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}} >, \mathbf{v} =< \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}} >$

Angle Between Vectors



$$\mathbf{u} =< u_1, u_2, u_3 >$$
$$\mathbf{v} =< v_1, v_2, v_3 >$$

Theorem: The angle θ between two vectors $\mathbf{u}$ and $\mathbf{v}$ is given by

$$\theta = \cos^{-1}\left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right).$$

Remark: We take θ in the range $0 \leq \theta \leq \pi$.

Remark: Dot products are also called *inner* or *scalar products* because the product results in a scalar, not a vector.

Example:

Find the angle between the vectors $\mathbf{u} = \sqrt{3}\mathbf{i} - 5\mathbf{j}$ and $\mathbf{v} = \sqrt{3}\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

Perpendicular (Orthogonal) Vectors

Two nonzero vectors **u** and **v** are **orthogonal** (or **perpendicular**) if and only if $\mathbf{u} \bullet \mathbf{v} = 0$.

Example: Suppose $P(1, 1, -1)$, $Q(0, 3, -2)$, $S(k, 0, 2)$ are points in space. Given that $\overrightarrow{PQ}$ is orthogonal to $\overrightarrow{PS}$, find k.

Dot Product Properties

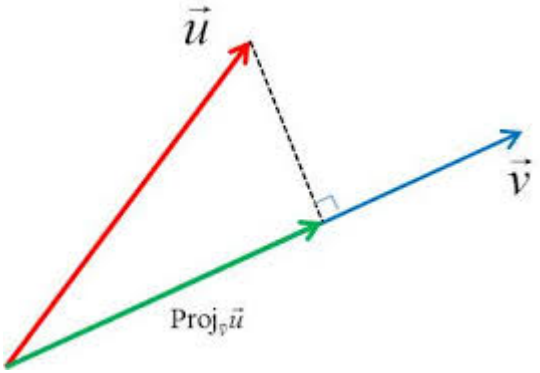
Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and let c be a scalar. Then,

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0$

Example:

Suppose $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors such that $|\mathbf{u}| = |\mathbf{v}|$. Show that $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal.

Vector Projections

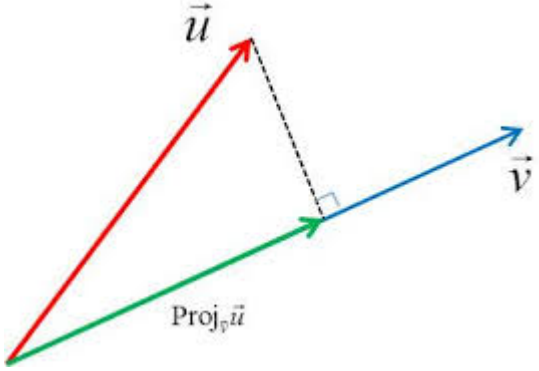


Drop the terminal point of $\mathbf{u}$ perpendicular to the line containing $\mathbf{v}$, we have

$$\text{Proj}_{\mathbf{v}}\mathbf{u}.$$

Example: Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$

Writing a Vector as a Sum of Orthogonal Vectors

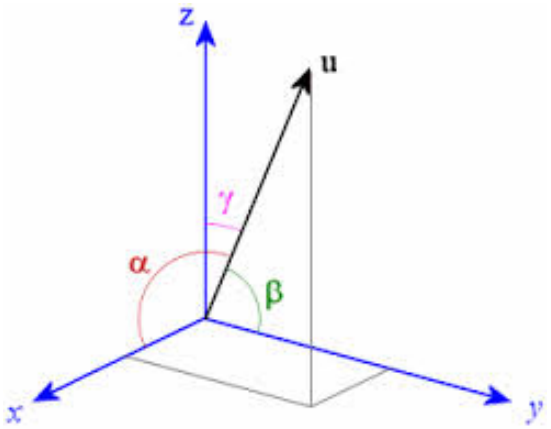


Example:

A force $\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ is applied to a spacecraft with velocity vector $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$. Express $\mathbf{F}$ as a sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

The Direction Cosines of Vectors in Space

Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, and let α, β, γ be the directed angles from the positive x-axis, y-axis, and z-axis, respectively, to the vector $\mathbf{u}$.



Then we have the following definitions.

- α, β, γ are called the *direction angles* of $\mathbf{u}$.
- (cos α, cos β, cos γ) are the *direction cosines* of $\mathbf{u}$.
- The ratios a: b: c are called the *direction ratios* of $\mathbf{u}$.

Theorem:

If α, β, γ are the direction angles of $\mathbf{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \neq \mathbf{0}$, then

1. $\cos\alpha = \frac{a}{|\mathbf{u}|}$, $\cos\beta = \frac{b}{|\mathbf{u}|}$ and $\cos\gamma = \frac{c}{|\mathbf{u}|}$.

2. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

Proof:

Example: Find the direction angles of $\mathbf{v} = \mathbf{i} + \mathbf{j} + \sqrt{2}\mathbf{k}$.

Remarks:

1. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma \neq 1 \Rightarrow$ does not exist vector with direction cosines $(\cos\alpha,\ \cos\beta,\ \cos\gamma)$.

2. $\mathbf{u}$ and $\mathbf{v}$ have the same direction cosines $\Leftrightarrow$ they have the same direction.

3. $\mathbf{u}$ and $\mathbf{v}$ have the same direction ratios $\Leftrightarrow$ they are parallel (i.e., they are in the same direction or in opposite direction)

4. Any vector $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \neq \mathbf{0}$ can be written as
$\mathbf{v} = |\mathbf{v}|(\cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k})$.

5. The unit vector $\mathbf{u}$ in the direction $\mathbf{v}$ is given by
$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = (\cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k})$.